

**Microprocessor Interfacing  
and Programming (EE3002)**

Date: November 4<sup>th</sup> 2024

Course Instructor(s)

Mr. Maaz Rizvi

Sessional-II Exam

Total Time (Hrs): 1

Total Marks: 15

Total Questions: 3

Roll No

Section

Student Signature

**Instructions**

Solve the paper on a separate answer sheet

You can use the Instruction Set provided to you on page 2 of the question paper

**Attempt all questions**

**Question No. 1 (CLO No. 3)**

Marks: 5

**Produce** a code in PIC Assembly to add four 8-bit numbers placed at the following locations. Store the higher byte and lower byte in registers named H\_BYTE and L\_BYTE respectively.

AB C D E F G H

H-BYTE L-BYTE

4C ---- 0x40

3F ---- 0x41

D3 ---- 0x42

A1 ---- 0x43

**Question No. 2 (CLO No. 3)**

Marks: 5

Assuming that XTAL is 16 MHz, **Produce** a code in PIC Assembly using **Timer 1** to generate a square wave of 40 ms on PORTB, 7. Show the calculations.

$$\frac{1}{4} \times 16 \times 10^6 = 4 \text{ MHz}$$

$$\frac{1}{4 \text{ MHz}} = 0.25 \mu$$

Marks: 5 (3+2)

**Question No. 3 (CLO No. 2)**

a)

**Explain** the role of Stack and subroutines in PIC. What happens when the Stack overflows?

b)

A developer is using PIC18 microcontroller for a product. The Chip has only 4K on-chip flash ROM. **Explain** which of the two instructions **CALL** or **RCALL** is more useful in programming this Chip.

SHRC  
B7

- ADDLW, Literal
- ANDLW, Literal
- SUBLW, Literal
- XORLW, Literal
- MULLW, Literal
- ADDWF fileReg, d
- ADDWFC fileReg, d
- ANDWF fileReg, d
- IORWF fileReg, d
- SUBFWB fileReg, d
- SUBWF fileReg, d
- SUBWFB fileReg, d
- XORWF fileReg, d
- COMF fileReg, d
- DECF fileReg, d
- DECFSZ fileReg, d
- DECFSNZ fileReg, d
- INCF fileReg, d
- INCFSZ fileReg, d
- INCSNZ fileReg, d
- MOVF fileReg, d
- NEGF fileReg, d
- RLCF fileReg, d
- RLNCF fileReg, d
- RRCF fileReg, d
- RRNCF fileReg, d
- SWAPF fileReg, d
- BTG fileReg, d
- IORLW, Literal
- MOVFF fileReg1, fileReg2
- BC
- BNC
- BZ
- BNZ
- BN
- BNC
- BOV
- BNOV
- BRA
- GOTO
- CALL
- RCALL
- RETURN
- NOP
- CLRF
- SETF
- BCF fileReg, bit
- BSF fileReg, bit
- BTFSS fileReg, bit

- BTFSC fileReg, bit
- BTG fileReg, bit
- STATUS
- C - Carry flag
- DC - Digital Carry flag
- Z - Zero flag
- OV - Overflow flag
- N - Negative flag
- CPFSGT fileReg, d
- CPFSEQ fileReg, d
- CPFSLT fileReg, d
- VAR\_NAME EQU Literal/Location
- ORG

- TMR0H/TMR1H
- TMR0L/TMR1L
- TMR0IF/TMR1IF
- INTCON
- PIR1

#### T0CON -

- TMR0ON D7 Timer0 ON and OFF control bit
- T08BIT D6 Timer0 8-bit/16-bit selector bit
- T0CS D5 Timer0 clock source select bit
- T0SE D4 Timer0 source edge select bit
- PSA D3 Timer0 prescaler assignment bit
- T0PS2:T0PS0 D2D1D0 Timer0 prescaler selector

#### T1CON -

- RD16 D7 16-bit read/write enable bit
- TICKPS2:TICKPS0 D5D4 Timer1 prescaler selector
- T1OSCEN D3 Timer1 oscillator enable bit
- T1SYNC D2 Timer1 synchronization
- TMR1CS D1 Timer1 clock source select bit
- TMR1ON D0 Timer1 ON and OFF con

MOVWF W  
20  
FU  
20  
21  
0A  
1